Security Implementation Strategy

Project: University Library Analytics Data Warehouse

1. Role-Based Access Control (RBAC)

To ensure that sensitive library data is only accessed by authorized personnel, we implemented RBAC in our Streamlit dashboard. The system dynamically filters SQL queries based on the logged-in user's role:

- **Admin:** Has full database access. Can view all dashboards, all departments, and export raw data.
- **Analyst:** Has read-only access to all dashboards for reporting purposes. Cannot modify the database.
- **Department Head:** Has restricted access. If the "Head of Engineering" logs in, the SQL queries append a `WHERE department = 'Engineering'` clause, ensuring they cannot see data from other departments.

2. Data Masking

To protect Personally Identifiable Information (PII), student IDs (e.g., STU-2024-001) are masked in the visual dashboards. When data is exported for external reporting, the system obfuscates the first half of the ID (e.g., showing only `****-2024-001`).

3. Encryption

- **In Transit:** The Streamlit application and the MySQL database are configured to use SSL/TLS encryption for all data transferred over the network, preventing man-in-the-middle attacks.
- **At Rest:** The MySQL database files are stored on an encrypted volume to ensure that if the physical hard drive is stolen, the data cannot be read.

4. Audit Logging

The database is configured with an audit log that tracks all user login attempts (successful and failed), as well as any INSERT, UPDATE, or DELETE operations. This log is reviewed weekly by the Admin to detect any suspicious activity or unauthorized access attempts.